

Lab 6

[valid 2024-2025]

Graphical User Interfaces (Swing, JavaFX)

Consider the following game played on a *drawing board* by two *players*.

At the beginning of the game, a number of dots (small circles) are randomly generated at various positions.

Each player, on their turn, chooses a pair of dots and connects them with a line (a player with blue, the other with red).

The goal of a player is to add enough lines such that all the dots and the lines form a connected structure (any two dots are connected by a path formed by lines and other dots) and to minimize the *score*, i.e. the sum of the lengths of the lines. The length of a line is the euclidean distance between its dots.

In order to create a graphical user interface for the game, you may use either Swing or JavaFX.

The main specifications of the application are:

Compulsory (1p)

Create the following components:

- The *main frame* of the application.
 - A *configuration panel* for introducing parameters, such as the number of dots, and a button for creating a new game. The panel must be placed at the *top* part of the frame. The panel must contain at least one label and one input component.
 - A *canvas (drawing panel)* for drawing the board. Draw the dots as small circles. This panel must be placed in the *center* part of the frame.
 - A *control panel* for managing the game. This panel will contain the buttons: *Load*, *Save*, *Exit*, etc. and it will be placed at the *bottom* part of the frame.
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Homework (2p)

- Implement the **retained** mode for drawing the game board.
 - Implement the logic of the game. You can add a line either by selecting two dots or by dragging a line from one dot to another.
 - Compare the score obtained by each player with the best score that could have been obtained.
 - Export the image of the game board into a PNG file.
 - Use object serialization in order to save and restore the current status of the game.
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Bonus (2p)

- Implement an algorithm that, considering the current status of the game, generates a number of connected structures (spanning trees) that could be created, descending by their score (cost), starting with the one with minimum cost.
- Implement an AI for the game. The algorithm will choose, when it is its turn to move, one of the spanning trees generated above, depending on a *difficulty level*. If the difficulty level is the highest, it will select the first generated tree, if it is the lowest, the last generated tree.
- When starting a new game, you should be able to choose for each player if it's human or AI, and select a difficulty level.

Resources

- [Slides](#)
- [Creating a GUI With JFC/Swing](#)
- [The Java Tutorials: 2D Graphics](#)
- [Getting Started with JavaFX](#)
- [JavaFX Scene Builder](#)

Objectives

- Get familiar with the basic elements of design involved in creating a GUI.
- Understand the concepts of *component*, *container*, *layout manager*.
- Get acquainted with various libraries for creating a GUI application, such as AWT, Swing, SWT, Java FX.
- Write *event listeners* to handle events.
- Understand how Swing components are *Painted*.
- Create custom components using Java2D that coexist with standard Swing components.
- Understand the concept of *graphic context*.
- Get familiar with Java2D basic *geometric* primitives, use *colours*, *fonts*, *images*
- Get acquainted with JavaFX technology and understand the differences between Swing and JavaFX.